

17 TF

①

a) 128. - → 191. - (first 2 bit : 10)

available network →

244

host per network : $2^{16} - 2$

| 1011111

A → 0 (18)

B → 10 (16)

C → 110 (12)

D → 1110

E → 1111

b) Nat preferred.

c)

	Circuit	datagram
speed	more	less (?)
overhead	more less	less more
utilization	less	more

d) For large → Link state.

Ovocal-D

Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX

Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

2)

a)

→ limit to a finite number of increase

→ horizon split

→ " " with poisoning

b)

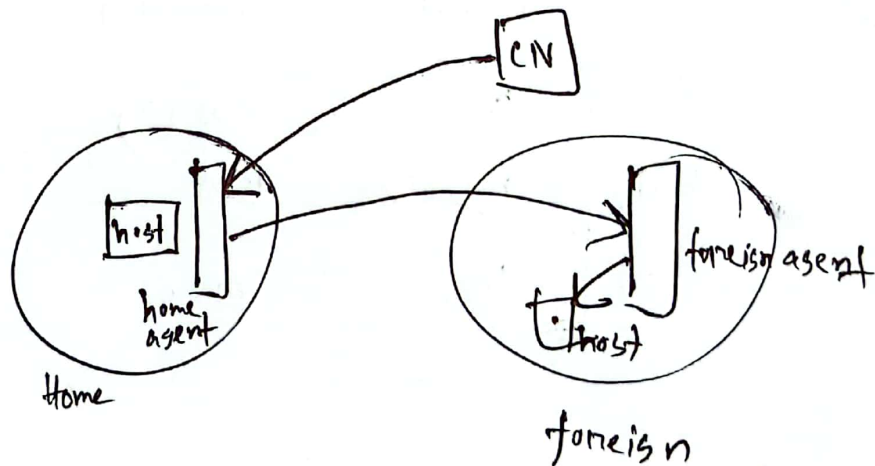
ABR → connect 2 areas on OSPF

ASBR → connects to the external (internet)

c)

objective:

continuous support (connectivity even when the host is in movement)



d) Slide 28 Page.

③

a) target → reliable connection (SYN, SYN-ACK, ACK)

exploit:

→ client sends a large SYN msg, but does not send the final ACK (3rd step). Server keeps waiting for these fraud connections and ignore legitimate connections (DOS)

b) Slide

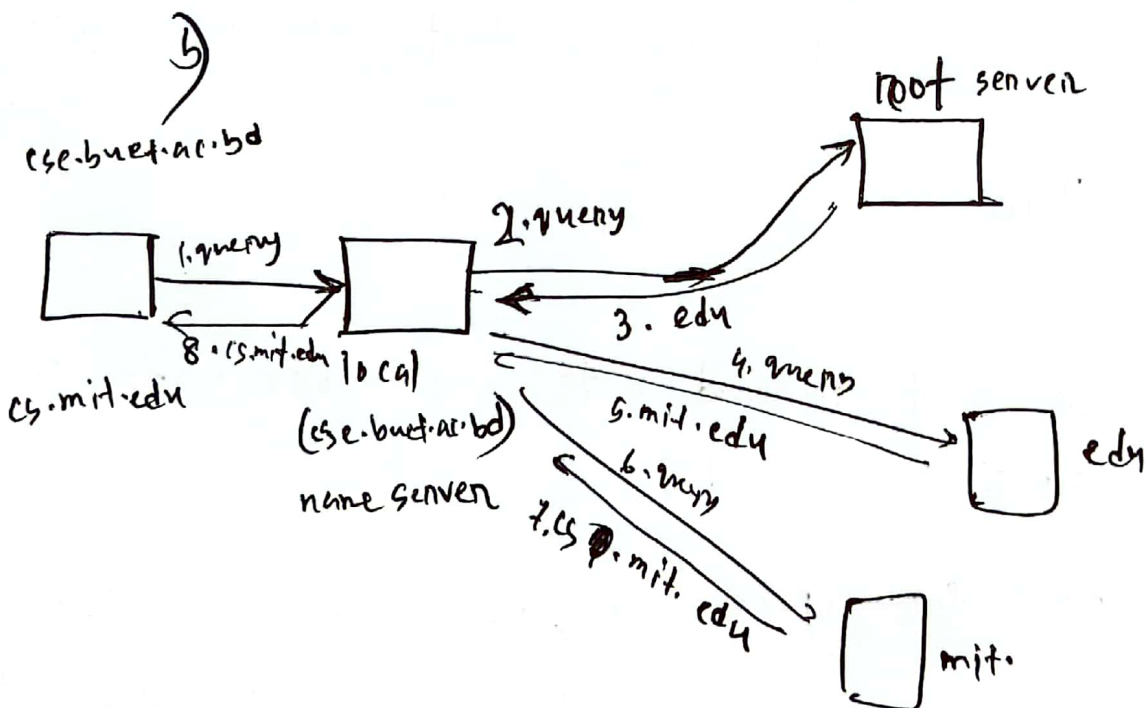
c) the last received byte was 4095, waiting for 4096 to onward data.

2K free space available to the receiver.

~~if~~ sender can send maximum 2K data starting from 4096.

④

- a) main diff: → POP3 deletes the mail as soon as read
 → IMAP creates a copy. if client/new user want to delete then removes it,
 → POP3: 110 port | imap: 143



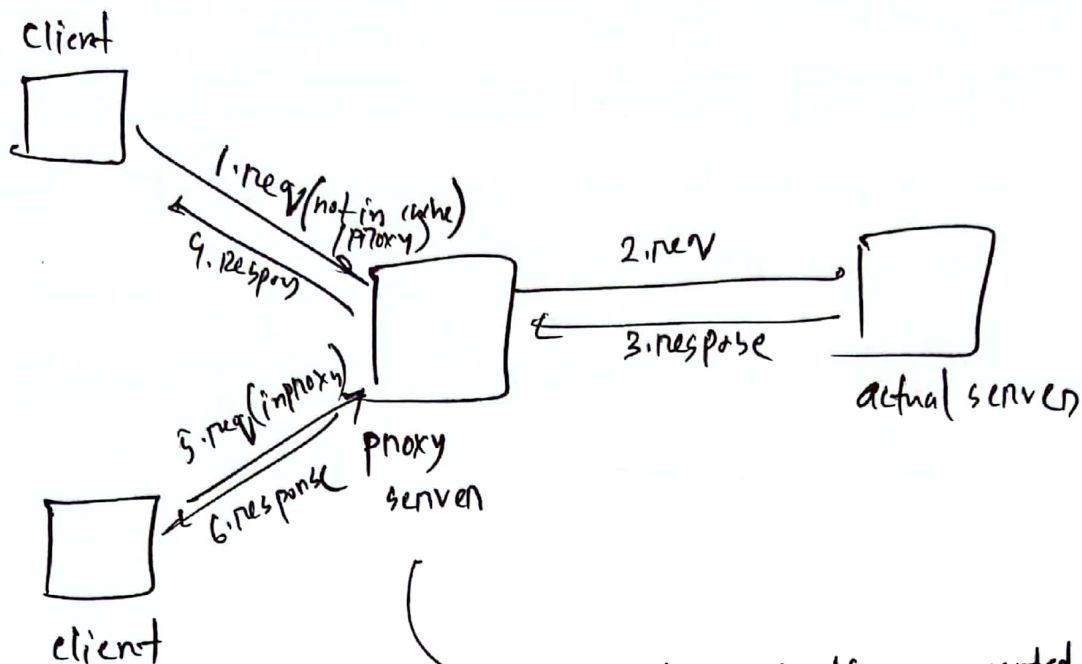
Ovocal-D

Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX

Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

C.



can monitor: traffic, unwanted sites
visit report, congestion,
security threats.

Section B

5

a) Not agree.

Reasons

1.

if the network traffic is low in
bit mapped scheme, the efficiency
is $= \frac{d}{d+N}$, d is data size, N is len of
contention slot which is a great overhead.
basic CSMA is better here.

Ovocal-D
Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

- ② if there are groups of stations, then better to use adaptive tree protocol. Moreover, among the groups we can assume that no collision happens generally, so we use Aloha inside a group. However, among the groups we can use bit mapped protocol as here collision can happen.



Using hub wont work. hub is physical layer device. it cannot control traffic flow among LANs based on the ~~net~~ LAN specifications.

Any promiscuous device will read data of all ~~the~~ clients,

→ Use bridge

→ the last part works with bridge.

Ovocal-D

Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX

Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

6

a) won't work.

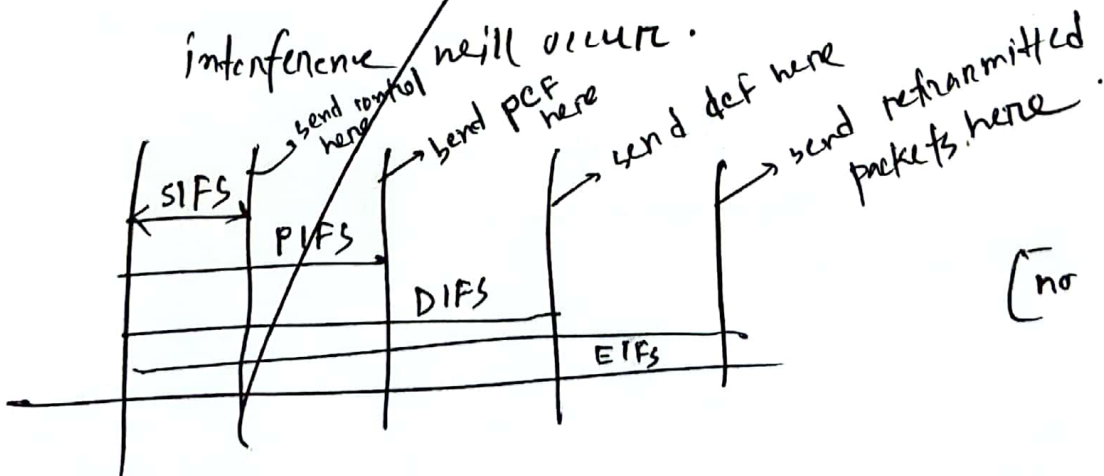
→ X consecutive codeword matrix can detect up to X length of burst error. 32 bit error won't be handled.

→ also, their count error other ideas are ok. count bit → 32 [12]

b)

Yes.

In a cell if we want to transmit point to point, or distributed coordination, along with control and retransmissions, it's essential to separate these frames, otherwise interference will occur.



[no interference]

Ovocal-D
Calcium 600 mg (eggshell source) & Vitamin D₃ 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D₃ 400 IU Tablet

~~Q~~

7

9) statement is valid $\rightarrow$ slide 2 $\rightarrow$ page 9

9) hybrid model:

- 1. physical
- 2. data link
- 3. network
- 4. transport
- 5. application.

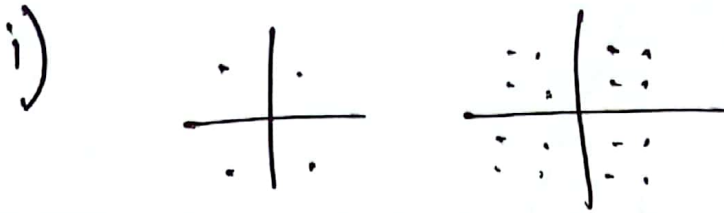
Ans to ans 8

~~Q~~

Ovocal-D
Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

a)



in QPSK $\rightarrow$ only phase (angle) changes

in QAM $\rightarrow$ phase (angle) and amplitude (len of vector) changes.

ii)

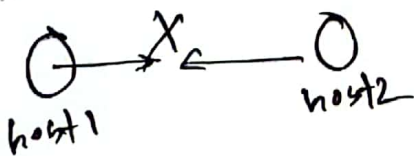
Geo $\rightarrow$ Geostationary earth orbit. 36,000 km.
1 day orbit time

LEO ~~LEO~~ $\rightarrow$ low earth orbit. below 2 km.
~~2-36,000 km~~

MEO $\rightarrow$ medium earth orbit. 2-36,000 km

b)

i)



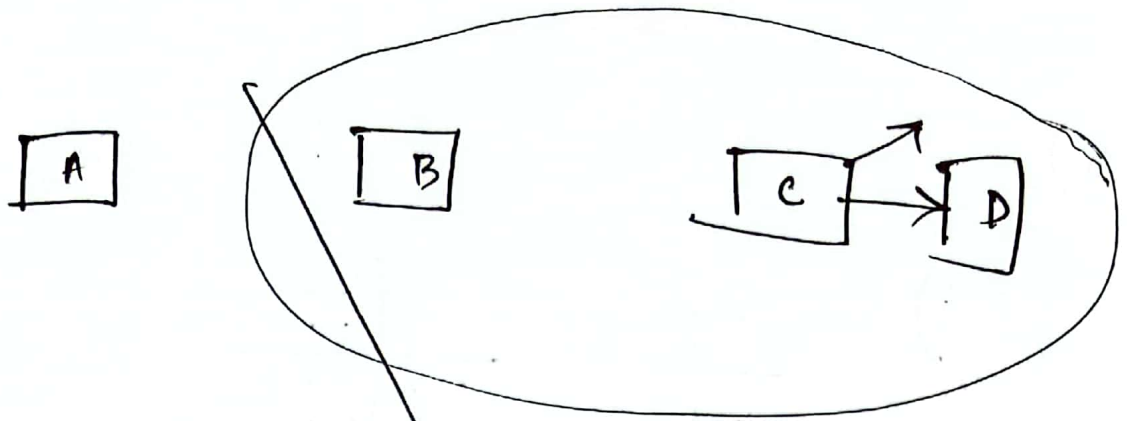
both tries to send data at same time
and col. occurs
 $\rightarrow$ for every ith collision chose a
number between (0 to 2ⁱ) and send
/retransmit after that time.

BEB

Ovocal-D
Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

ii)



B want to send to A. But in B's range C is transmitting. B might think, C is transmitting to A. so he won't transmit. However it was actually free.

ethernet
(wired)
CSMA/CD issue

16 batch

(A)

(1)

(2)

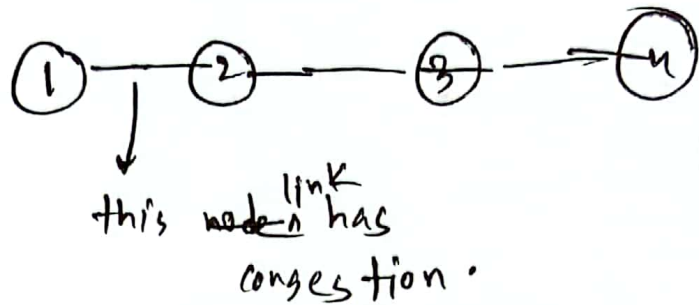
17 2.16.10. 0/23.

	host number	range	subnet
100	7	10.0 - 10.127	255.255.255.128
90	7	10.128 - 10.255.	" .128
30	5	11.0 - 11.31	" .224
12	4	11.32 - 11.47	" .240
10	4	11.48 - 11.63.	" .240

Ovocal-D
Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

2a) agree reason →



if hop by hop, 1 → 2 this hop will resolve it.

but if full traverse, then need to traverse the full path.

b) slide

c) prev gen.

3a) prev slide

b) describe.

c) buffer is full, wait to read by the receiver.

4

a)

DNS resolution by local DNS server. Local server is not the actual parent, doesn't have real zone file. Only have a cache.

→ Not really. The cache can provide false data if intended to cause privacy harm.

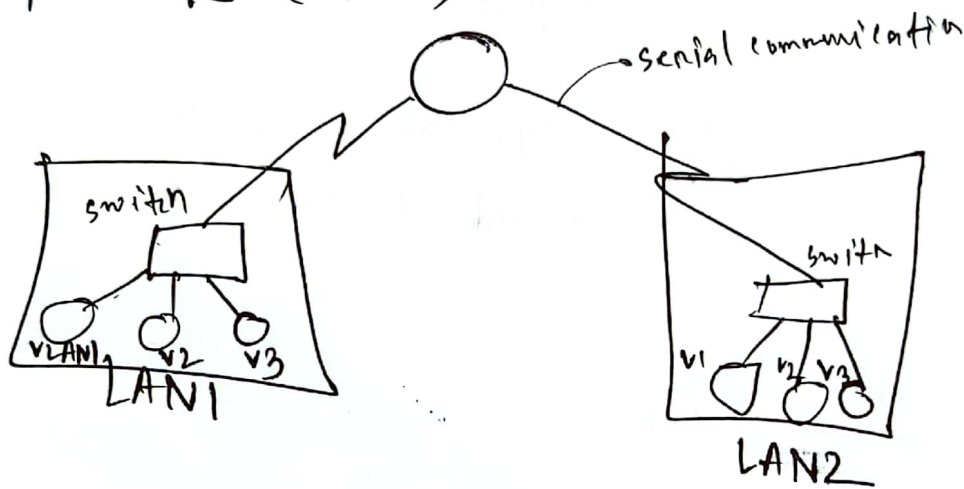
b) slide

c) slide

B

5)

a) possible. (not sure)



b)

Hidden/ exposed these are related to wireless communication.

but wdma is used in fiber optics which is wired. So not possible.

Ovocal-D

Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX

Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

6

a/

nope.

reason $\rightarrow$ timeout concept is both in Grobuck N
and selective repeat.

b/

x2

x4

x6

x5

x7.



Ovocal-D
Calcium 500 mg (eggshell source) & Vitamin D₃ 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D₃ 400 IU Tablet

1

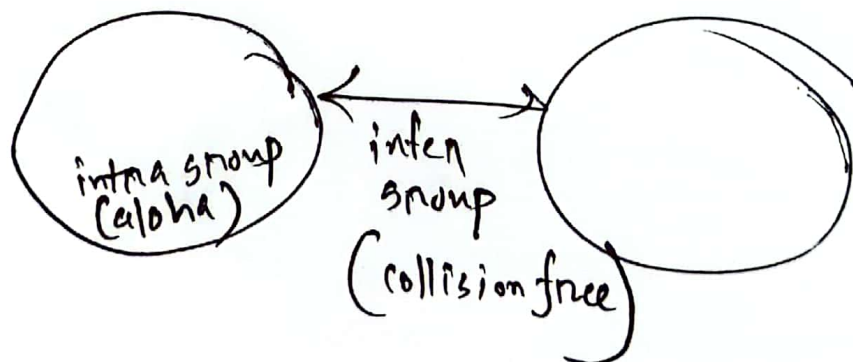
a) No, अलग concept

b) yes

→ explain, describe



a)



Ovocal-D
Calcium 500 mg (eggshell source) & Vitamin D, 200 IU Tablet

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D, 400 IU Tablet

b

not possible. totally unrelated.

→ order not maintained.

Ovocal-D

Ovocal-DX
Calcium 600 mg (eggshell source) & Vitamin D₃ 400 IU Tablet